

The Complex Field

Complex Analysis

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Definition

A group $\mathcal{G}$ is a set of elements $\{X, Y, \dots\}$, together with a rule for combining them that associates with each ordered pair X, Y a 'product' or combination law $X \bullet Y$ for which the following conditions must be satisfied.

- i) For every pair of elements X, Y that belongs to $\mathcal{G}$, the product $X \bullet Y$ also belongs to $\mathcal{G}$. (This is known as the **closure property**)
- ii) For all triples X, Y, Z the **associative law**; in symbols,

$$X \bullet (Y \bullet Z) = (X \bullet Y) \bullet Z.$$

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iii) *There exists a unique element I , belonging to $\mathcal{G}$, with the property that*

$$I \bullet X = X = X \bullet I$$

*for all X belonging to $\mathcal{G}$. This element is known as the **identity element** of the group.*

iv) *For every element X of $\mathcal{G}$, there exists an element X^{-1} , also belonging to $\mathcal{G}$, such that*

$$X^{-1} \bullet X = I = X \bullet X^{-1}.$$

*X^{-1} is called the **inverse** of X .*

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Theorem

The identity of any particular group and the inverse of any particular element is unique.

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Theorem

The inverse of a product of elements in a group is the product of the inverses in reverse order.

Theorem

Given any pair of elements X, Y belonging to $\mathcal{G}$, there exist unique elements U, V , also belonging to $\mathcal{G}$, such that

$$X \bullet U = Y \quad \text{and} \quad V \bullet X = Y.$$

Theorem (Cancellation Law)

The **cancellation law** can be stated as follows. If

$$X \bullet Y = X \bullet Z$$

for some X belonging to $\mathcal{G}$, then $Y = Z$. Similarly,

$$Y \bullet X = Z \bullet X$$

implies the same conclusion.

Definition

If any two particular elements of a group satisfy

$$X \bullet Y = Y \bullet X,$$

*they are said to **commute** under the operation $\bullet$; if all pairs of elements in a group satisfy such condition, then the group is said to be **Abelian**.*

Definition

Suppose $\mathcal{R}$ is a set in which a relation of equality, denoted by $=$, and operations of addition and multiplication, denoted by $+$ and $\cdot$, respectively, are defined. Then $\mathcal{R}$ is a **ring** (with respect to these operations) if these conditions hold:

- i) $\mathcal{R}$ forms an Abelian group with respect to addition.
- ii) $\mathcal{R}$ is closed with respect to an associative multiplication.
- iii) Two distributive laws hold in $\mathcal{R}$: $x \cdot (y + z) = x \cdot y + x \cdot z$ and $(x + y) \cdot z = x \cdot z + y \cdot z$ for all x, y, z in $\mathcal{R}$.

Definition

Let $\mathcal{R}$ be a ring. Then the additive identity of $\mathcal{R}$ is denoted by 0 and referred to as the **zero** of $\mathcal{R}$ and has the defining property $x + 0 = x$ for all $x \in \mathcal{R}$. If $y \in \mathcal{R}$, then the additive inverse $-y$ is called the **negative** of y or the **opposite** of y and has the defining property $y + (-y) = 0$. Furthermore, if $x, y \in \mathcal{R}$, then the **subtraction** in $\mathcal{R}$ is defined by

$$x - y := x + (-y).$$

Definition

*Let $\mathcal{R}$ be a ring. If there exists an element e in $\mathcal{R}$ such that $x \cdot e = e \cdot x = x$ for all x in $\mathcal{R}$, then e is called a **unity**, and $\mathcal{R}$ is said to be a **ring with unity**. If multiplication in $\mathcal{R}$ is commutative, then $\mathcal{R}$ is called a **commutative ring**.*

Definition

Let $\mathcal{R}$ be a ring with unity e , and let $a \in \mathcal{R}$. If there is an element x in $\mathcal{R}$ such that $ax = xa = e$, then x is a **multiplicative inverse** of a and a is called a **unit** (or an **invertible element**) in $\mathcal{R}$.

Definition

Let $\mathbb{F}$ be a ring. Then $\mathbb{F}$ is said to be a **field** if the following conditions hold.

- i) $\mathbb{F}$ is a commutative ring.
- ii) $\mathbb{F}$ has a unity e , and $e \neq 0$.
- iii) Every nonzero element of $\mathbb{F}$ has a multiplicative inverse.

Definition

A **complex number** is any number of the form

$$z = a + ib$$

where a and b are real numbers and i is the **imaginary unit**, defined by the property $i^2 = -1$. Here, we say that a is the **real part** of z ; b is called the **imaginary part** of z , denoted by $\Re(z)$ and $\Im(z)$, respectively.

Definition

*Two complex numbers z_1 and z_2 are said to be **equal** if and only if $\Re(z_1) = \Re(z_2)$ and $\Im(z_1) = \Im(z_2)$.*

Definition

If z_1, z_2 are complex numbers, we define:

a) $z_1 + z_2 = \Re(z_1) + \Re(z_2) + i(\Im(z_1) + \Im(z_2)).$

b) $z_1 - z_2 = \Re(z_1) - \Re(z_2) + i(\Im(z_1) - \Im(z_2)).$

c) $z_1 \cdot z_2 = \Re(z_1)\Re(z_2) - \Im(z_1)\Im(z_2) + i(\Re(z_1)\Im(z_2) + \Im(z_1)\Re(z_2)).$

Definition

If z is a complex number, the number obtained by changing the sign of its imaginary part is called the **complex conjugate**, or simply **conjugate**, of z and is denoted by the symbol $\bar{z}$. In other words, if $z = a + ib$, then its conjugate is $\bar{z} = a - ib$.

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*If z is a complex number, then the **modulus** of z is denoted by $|z|$ and defined as the positive square root of the number $z\bar{z}$.*

Definition

If z_1, z_2 are complex numbers, we define:

$$\frac{z_1}{z_2} = z_1 z_2^{-1} = \frac{z_1 \bar{z}_2}{|z_2|^2}.$$

Definition

The **zero** in the complex numbers is the number $0 + 0i$ and the **unity** is $1 + 0i$. The zero and unity are denoted by 0 and 1 , respectively. The zero is the additive identity in the complex numbers since, $z + 0 = z$. Similarly, the unity is the multiplicative identity since, $z \cdot 1 = z$.

Proposition

The set

$$\mathbb{C} := \{a + ib : a, b \in \mathbb{R}, i^2 = -1\},$$

along with the previously defined relation of equality ($=$) and operations $(+, \cdot)$ forms a field (with respect to these operations).

*We will refer to $\mathbb{C}$ as the **complex field**.*